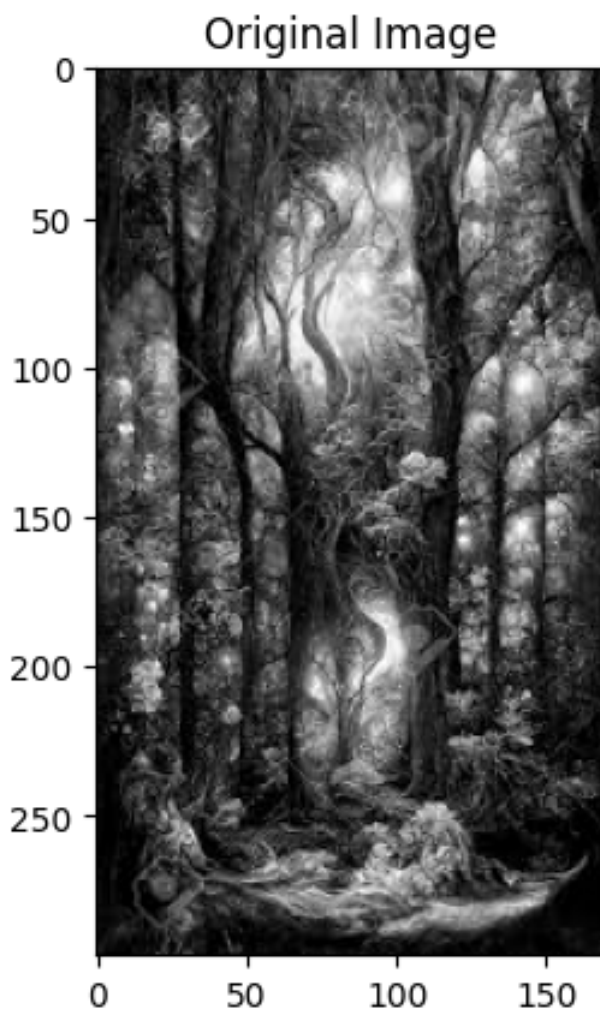


✓ Task 1

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
```

```
img = cv2.imread('images.jpg', 0)
plt.imshow(img, cmap='gray')
plt.title("Original Image")
plt.show()
```

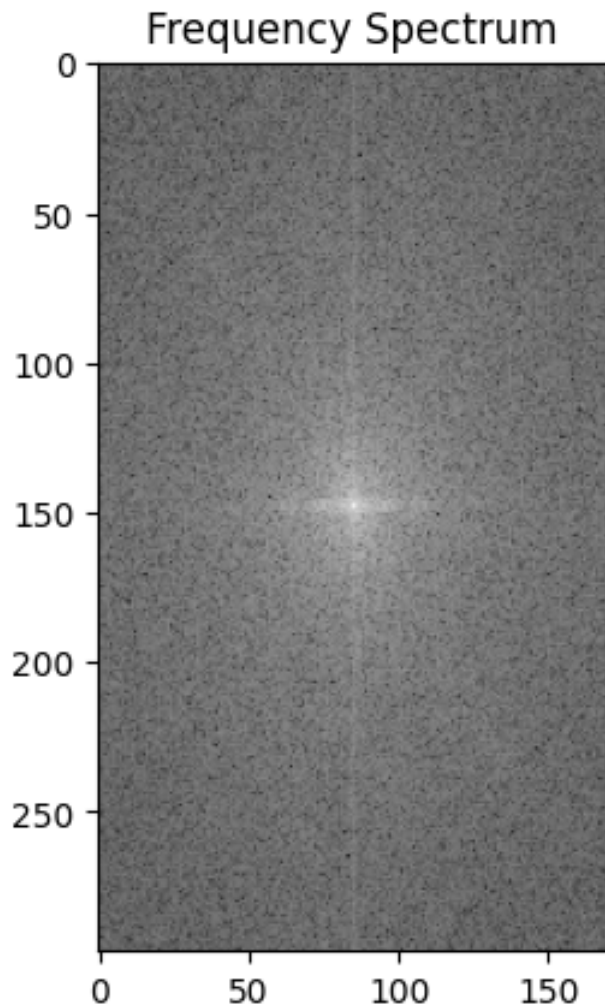


```
f = np.fft.fft2(img)
fshift = np.fft.fftshift(f)
```

```

magnitude_spectrum = 20*np.log(np.abs(fshift))
plt.imshow(magnitude_spectrum, cmap='gray')
plt.title("Frequency Spectrum")
plt.show()

```



```

rows, cols = img.shape
crow, ccol = rows//2 , cols//2
mask = np.zeros((rows, cols), np.uint8)
mask[crow-30:crow+30, ccol-30:ccol+30] = 1
fshift_filtered = fshift * mask

```

```

f_ishift = np.fft.ifftshift(fshift_filtered)
img_back = np.fft.ifft2(f_ishift)
img_back = np.abs(img_back)

```

```

mask = np.ones((rows, cols), np.uint8)
mask[crow-30:crow+30, ccol-30:ccol+30] = 0
fshift_filtered = fshift * mask

```

```
f_ishift = np.fft.ifftshift(fshift_filtered)
img_back = np.fft.ifft2(f_ishift)
img_back = np.abs(img_back)
```

✓ Task 2

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
```

```
img = cv2.imread('images.jpg')
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
gray = np.float32(gray)
```

```
dst = cv2.cornerHarris(gray, 2, 3, 0.04)
```

```
dst = cv2.dilate(dst, None)
```

```
img[dst > 0.01 * dst.max()] = [0, 0, 255]
```

```
plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
plt.title("Harris Corners")
plt.show()
```

Harris Corners

